

MUMBLES HEAD, United Kingdom

WMO: 036090

Lat: 51.5656N

Lon: 3.9817W

Elev: 43

StdP: 100.81

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	0.9	-5.0	2.5	1.5	-3.6	2.8	2.5	27.5	8.6	24.7	8.7	6.3	30	0.750

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	22.6	17.8	21.2	17.1	20.0	16.6	18.5	21.1	17.8	20.0	17.2	19.1	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.7	19.2	17.0	12.2	18.6	16.5	11.8	18.2	52.4	21.3	50.4	20.1	48.7	19.1	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	18.1	15.2	DB	-1.3	25.4	1.6	1.8	-2.5	26.7	-3.4	27.7	-4.3	28.8	-5.5	30.1	(4)
(5)			WB	-2.8	19.9	1.5	1.1	-3.9	20.7	-4.8	21.3	-5.6	21.9	-6.7	22.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	6.5	6.1	7.3	9.6	12.1	14.9	16.9	16.8	15.2	12.8	9.6	7.2	(6)	
	DBStd	4.45	2.38	2.35	2.14	2.21	2.11	2.05	1.99	1.72	1.77	2.21	2.51	2.74	(7)	
	HDD10.0	468	107	108	85	33	4	0	0	0	0	6	36	89	(8)	
	HDD18.3	2592	366	341	341	263	192	105	55	54	96	172	263	344	(9)	
	CDD10.0	938	0	0	2	19	70	148	214	211	155	91	23	4	(10)	
	CDD18.3	21	0	0	0	0	0	3	10	7	1	0	0	0	(11)	
	CDH23.3	31	0	0	0	0	0	6	14	11	0	0	0	0	(12)	
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0	(13)	
(14) Wind	WSAvg	7.0	7.9	7.3	6.7	6.2	6.6	6.3	6.2	6.5	6.6	7.8	7.4	7.9	(14)	
(15) Precipitation	PrecAvg	1198	131	86	92	71	71	76	70	99	107	127	139	135	(15)	
	PrecMax	1542	255	204	226	169	149	216	203	242	240	281	300	325	(16)	
	PrecMin	862	8	0	13	6	6	10	9	7	23	18	64	34	(17)	
	PrecStd	166	62	49	50	38	37	46	40	56	47	60	54	65	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.4	13.9	17.9	20.8	23.7	25.1	24.7	21.4	18.3	14.8	13.0	(19)	
		MCWB	10.3	9.2	9.9	13.0	16.0	17.5	18.6	19.2	17.6	15.8	13.4	11.9	(20)	
	2%	DB	10.6	10.4	12.0	15.6	18.4	21.2	22.9	22.1	19.9	17.1	14.1	12.1	(21)	
		MCWB	9.8	9.0	9.2	11.7	14.4	16.2	17.9	18.0	16.6	15.0	12.9	11.2	(22)	
	5%	DB	10.1	9.7	10.9	13.9	16.7	19.5	21.5	20.6	18.8	16.3	13.5	11.6	(23)	
		MCWB	9.3	8.6	8.7	10.9	13.6	15.7	17.3	17.1	15.9	14.6	12.4	10.7	(24)	
	10%	DB	9.6	9.1	10.1	12.6	15.3	18.1	20.1	19.5	17.9	15.7	13.0	10.9	(25)	
		MCWB	8.8	8.1	8.5	10.1	12.7	15.0	16.7	16.6	15.5	14.1	11.8	10.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.2	10.7	13.8	16.8	18.7	19.5	20.1	18.2	16.5	14.1	12.2	(27)	
		MCDB	10.8	10.5	12.6	16.0	19.5	22.2	23.3	23.1	20.1	17.7	14.3	12.5	(28)	
	2%	WB	10.1	9.6	9.9	12.3	15.1	17.1	18.5	18.5	17.4	15.7	13.3	11.6	(29)	
		MCDB	10.4	9.9	11.1	14.8	17.3	19.9	21.5	20.8	18.9	16.5	13.7	12.0	(30)	
	5%	WB	9.6	9.0	9.4	11.3	14.1	16.2	17.8	17.8	16.8	15.1	12.7	11.0	(31)	
		MCDB	9.9	9.4	10.4	13.3	16.0	18.6	20.4	19.6	18.1	15.9	13.3	11.4	(32)	
	10%	WB	9.1	8.4	8.9	10.5	13.2	15.5	17.2	17.2	16.2	14.5	12.1	10.2	(33)	
		MCDB	9.5	8.9	9.7	12.1	14.8	17.5	19.4	18.8	17.5	15.5	12.8	10.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	3.6	4.2	4.9	4.9	4.9	4.9	4.7	4.7	4.0	3.7	3.4	(35)	
		MCDBR	3.3	3.6	4.9	6.3	6.7	7.0	6.9	6.3	5.7	4.0	3.4	3.4	(36)	
	5% WB	MCWBR	3.6	3.2	3.6	4.0	4.2	4.1	3.9	3.9	3.8	3.2	3.3	3.7	(37)	
		MCWBR	3.3	3.2	4.1	5.5	5.9	5.7	5.1	4.5	3.4	3.2	3.2	3.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.331	0.343	0.372	0.388	0.395	0.385	0.386	0.371	0.364	0.353	0.329	0.320	(40)	
		taud	2.350	2.389	2.323	2.305	2.327	2.386	2.387	2.445	2.454	2.448	2.407	2.325	(41)	
	Ebn at Noon	Ebn at Noon	670	776	818	849	859	870	862	859	822	755	670	612	(42)	
		Edh at Noon	64	80	102	117	120	114	112	101	89	75	60	56	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.76	1.50	2.68	4.21	5.16	5.61	5.18	4.31	3.22	1.80	0.93	0.59	(44)	
	RadStd	RadStd	0.07	0.16	0.29	0.39	0.29	0.53	0.46	0.33	0.24	0.15	0.08	0.07	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page